LEARNER GUIDE

The circle and its parts

Circle Theorems · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A circle looks plain, but it hides a whole vocabulary, and every circle theorem to come is built from these words. The curve itself is the **circumference**; its middle is the **centre**. A straight line from the centre to the curve is a **radius**; a straight line right across through the centre is a **diameter** (two radii in a line, and the longest chord of all). Any straight line joining two points on the curve is a **chord**. A piece of the curve is an **arc**. A line that grazes the circle at exactly one point is a **tangent**. The curve also cuts the inside into **regions**: a wedge between two radii is a **sector**; the region between a chord and its arc is a **segment**. Finally, an angle such as ∠APB is said to **stand on** (or **subtend**) the arc AB that does **not** contain P — the single idea the whole unit turns on.

What you are learning to notice

- Sort the parts into families: **lines** (radius, diameter, chord, tangent), **curves** (arc, circumference) and **regions** (sector, segment).
- A **diameter** is a chord through the centre — and the **longest** chord there is.
- A **sector** is bounded by two radii; a **segment** by one chord. That is what tells them apart.
- An angle **stands on** the arc on the far side from its vertex — the arc that does not contain it.

Moving through it, one step at a time

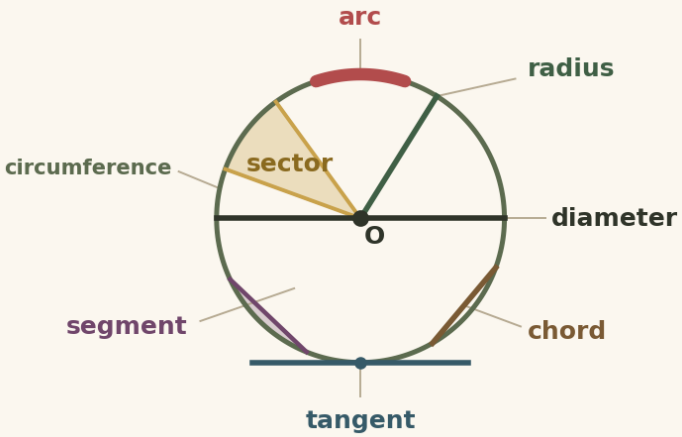
To name a part, ask what bounds it. From the centre to the curve → radius. Through the centre, right across → diameter. Two points on the curve, not through the centre → chord. Touching at one point → tangent. A piece of the curve → arc. Two radii and an arc → sector. One chord and an arc → segment. For an angle ∠APB on the circle, the arc it stands on is the arc AB that does not contain P.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Name the parts of a circle

Identify each labelled feature and say whether it is a line, a curve or a region.

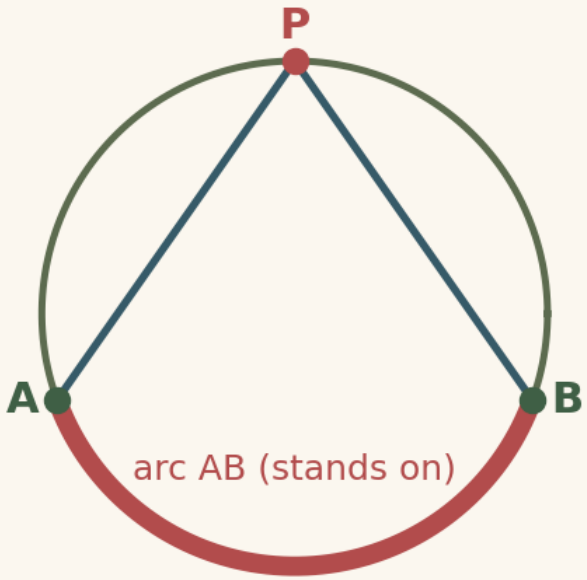


1. From the centre O out to the curve — one straight line.	radius
2. Right across through O — the longest chord.	diameter
3. Two points on the curve, not through O.	chord (a line); the curve itself is the circumference
4. A wedge between two radii, and the cap between a chord and its arc.	sector (region) and segment (region)

Lines: radius, diameter, chord, tangent. Curves: arc, circumference. Regions: sector, segment. The centre O is a single point.

Example 2 — Which arc does the angle stand on?

Points A, B and P lie on the circle. On which arc does ∠APB stand?



1. Find the vertex of the angle — the point at the corner.	the vertex is P
2. The angle stands on the arc AB on the FAR side from P.	the arc not containing P
3. Slide P along its own arc — the arc it stands on does not change.	still the same arc AB

∠APB stands on (subtends) the arc AB that does not contain P — here the lower, major arc. The same arc AB also subtends an angle at the centre, ∠AOB, which the next lesson connects to it.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (knowing a circle as a round curve, angle notation such as ∠APB, and the idea of a straight line joining two points), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.

- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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